# Practice workbook · Subprograms and strings: complete the function and its call

A short Paper 2 program built around subprograms and string handling: completing a function or procedure body, passing arguments, returning a value, calling it from the main program or a menu, and string work inside it (slicing, joining with numbers, reversing, checking length with re-input).

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: CSC-CODE-07

# 1. Question 4 · part c · 6 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

**4** Programmers write code that keeps track of values, is efficient and uses subprograms.

(a) Global variables are different from local variables.

(i) State where a **global** variable is created.

**(1)**

(ii) State where a **local** variable is accessible.

**(1)**

(b) An algorithm reports the range that an integer falls within.

Integers from 0 to 39 are in the low range.

Integers from 40 to 69 are in the middle range.

Integers from 70 to 100 are in the high range.

**Figure 4** and **Figure 5** show two algorithms that solve this problem.

```
1  SEND 'Enter an integer: ' TO DISPLAY
2  RECEIVE theNumber FROM (INTEGER) KEYBOARD
3  IF (theNumber < 40) THEN
4      SEND 'Low range' TO DISPLAY
5  ELSE
6      IF (theNumber < 70) THEN
7          SEND 'Middle range' TO DISPLAY
8      ELSE
9          SEND 'High range' TO DISPLAY
10     END IF
11 END IF
```

**Figure 4**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8

P 7 5 7 3 7 A 0 8 1 6

# 1. Question 4 · part c · 6 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

```
1   SEND 'Enter an integer: ' TO DISPLAY
2   RECEIVE theNumber FROM (INTEGER) KEYBOARD
3   IF (theNumber < 40) THEN
4       SEND 'Low range' TO DISPLAY
5   END IF
6   IF (theNumber > 39) AND (theNumber < 70) THEN
7       SEND 'Middle range' TO DISPLAY
8   END IF
9   IF (theNumber > 69) THEN
10      SEND 'High range' TO DISPLAY
11  END IF
```

**Figure 5**

Explain the reason that the algorithm in **Figure 4** is more efficient than the algorithm in **Figure 5**.

**(2)**

OLevels.com SCHOOL

# 1. Question 4 · part c · 6 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

(c) A program is required to create a new key.

The program takes two inputs.

The first input is a four-character string.

The second input is a whole number.

The key is constructed by joining the first two characters from the string, the number and the final two characters from the string.

When the user enters the four-character string **abcd** and the integer **123**, the program must construct and display the new key **ab123cd**

Open **Q04c** in the code editor.

Amend the code to:

- complete the subprogram to construct the new key
- complete the call to the subprogram.

Do not add any further functionality.

Save your code as **Q04cFINISHED** with the correct file extension for the programming language.

**(6)**

**(Total for Question 4 = 10 marks)**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

## 1. Question 4 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

## 2. Question 4 · part b · 7 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

**4** Gianluca is a schoolteacher. He is in charge of a year group of pupils.

(a) He has started to create a program to analyse test results.

Open **Q04a** in the code editor.

Amend the code to complete the 'if statement' used to produce the outputs described in this table.

| Condition | Output text |
|---|---|
| English test score was less than 40;<br>Maths test score was less than 50 | Student failed both tests |
| English test score was less than 40 or<br>Maths test score was less than 50 | Student failed one test |
| English test score was at least 80;<br>Maths test score was at least 85 | Student passed both tests with distinction |
| Other conditions | Student passed both tests |

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

**(6)**

(b) Gianluca has started to create a program to analyse pupil attendance.

He wants to use a menu system with subprograms.

Open **Q04b** in the code editor.

The program must include subprograms to:

- display the **names** of students whose percentage attendance was less than 75%
- count and display the **number** of students whose percentage attendance was 90% or higher.

Complete the program code to implement these requirements using the structure given in **Q04b**.

Do not add any further functionality.

Save your code as **Q04bFINISHED** with the correct file extension for the programming language.

**(7)**

**(Total for Question 4 = 13 marks)**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AR

OLevels.COM SCHOOL

## 2. Question 4 · part b · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

## 3. Question 4 · part a · 7 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

**4** Zak sells snacks at an after-school club.

(a) He wants a program that will hold details of the products he sells.

The program must:

- generate a five-character product code that:
  - starts with the first three letters of the product name
  - ends with a random number between 10 and 30
- display the product code followed by the product name.

Open **Q04a** in the code editor.

Write the program.

You must use the structure given in the file **Q04a** to complete the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

**(7)**

(b) Zak plans to implement a binary search algorithm to search a table of products.

(i) Explain **one** advantage of a binary search compared to a linear search.

**(2)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

## 3. Question 4 · part a · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

# 4. Question 4 · part a · 11 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

**4** Programs can be used to manipulate data.

(a) A program is needed to convert an 8-bit binary number into denary.

The program must:

- receive the input of a binary pattern and validate that it has exactly eight characters (there is no need to validate that it only contains 0s and 1s)
- request the input again if the binary pattern does not have exactly eight characters
- when a valid binary pattern is input:
  - convert the binary pattern into its denary equivalent
  - print a message that contains the binary pattern and its denary equivalent
- include a comment to explain how the conversion process works.

HINT

| Denary placeholder | 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 |
|---|---|---|---|---|---|---|---|---|
| Binary number | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| Denary number | 128 | 0 | 32 | 16 | 0 | 4 | 0 | 0 |

128 + 32 + 16 + 4 = 180

Open **Q04a** in the code editor.

Write a program to implement these requirements.

You must use the structure given in **Q04a** to write the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

**(11)**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

10

P 6 9 2 9 8 A 0 1 0 1 6

# 4. Question 4 · part a · 11 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Open **Q04b** in the code editor.

Describe the purpose of the program.

**(2)**

**(Total for Question 4 = 13 marks)**

OLevels.com SCHOOL

## 4. Question 4 · part a · 11 marks · Working and answer

Working and answer

OLevels.com SCHOOL

# 5. Question 4 · part b · 8 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

**4** Data is encoded for encryption and for identifying records in databases.

(a) A Caesar cipher is one method of encryption.

(i) Complete the table to show the result of applying a Caesar cipher.

**(2)**

| Plaintext | Shift | Ciphertext |
|---|---|---|
| PIXEL | −4 | |
| CLOUD | | FORXG |

(ii) Plaintext is encrypted using a Caesar cipher.

Compare the ciphertexts produced for the plaintext **GOLD** by:

- a shift of −6 followed by a shift of +8
- a single shift of +2.

**(2)**

OLevels.com SCHOOL

(iii) Ann has written a Caesar cipher algorithm.

When the algorithm encrypts the plaintext **BYTE** with a shift of +5 the result is **GVYJ**.

This is incorrect.

Explain the error.

**(2)**

## 5. Question 4 · part b · 8 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

(b) A program is needed to create a key for a database.

A user enters a two-letter word, a whole number and a decimal number.

The program must ensure the word is only two characters long.

The program must display an error message when the word is not two characters long.

A key is generated from the whole number, the reversed word and the whole number part of the decimal number.

**Figure 4** shows the input values and a valid key.

```
Enter a word: at
Enter a whole number: 12
Enter a decimal number: 7.89
12ta7
```

**Figure 4**

Open **Q04b** in the code editor.

Write the program.

Do not add any further functionality.

Save your code as **Q04bFINISHED** with the correct file extension for the programming language.

(8)

**(Total for Question 4 = 14 marks)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

## 5. Question 4 · part b · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

# 6. Question 2 · part c.ii · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

**2** Programs can process numerical data.

(a) Identify the method used to check input data matches set rules.

**(1)**

- ☒ **A** Abstraction
- ☒ **B** Encryption
- ☒ **C** Iteration
- ☒ **D** Validation

(b) **Figure 2** shows a section from an employee payslip produced by a computer program.

| Pay Rate | Hours | | Total Pay |
|---|---|---|---|
| £ 18.75 | 19 | | £ 356.25 |
| | Tax | 20% | £ 71.25 |
| | | Income | £ 285.00 |

**Figure 2**

OLevels.COM SCHOOL

(i) Complete the table to give **one** data type shown in **Figure 2** and give an example value.

**(2)**

| | |
|---|---|
| **Data type** | |
| **Example** | |

(ii) Complete the table to give an input, a process and an output used in the program to produce the payslip.

**(3)**

| | |
|---|---|
| **Input** | |
| **Process** | |
| **Output** | |

# 6. Question 2 · part c.ii · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part c.ii only.

(c) Numerical values can be represented in denary, binary and hexadecimal.

A program converts denary values in the range 0 to 255 into hexadecimal.

(i) The program is tested using appropriate integer values.

Complete the table to show an example of normal, boundary and erroneous test data for this program.

(2)

| | Test data |
|---|---|
| **Normal** | |
| **Boundary** | |
| **Erroneous** | |

(ii) The program uses subprograms to do the conversion.

Open **Q02cii** in the code editor.

Amend the code to complete the program to:

- convert the user input to integer
- check the input is between 0 and 255
- call the **getHex** subprogram, with the user input as an argument.

Do not add any further functionality.

Save your code as **Q02ciiFINISHED** with the correct file extension for the programming language.

(4)

**(Total for Question 2 = 12 marks)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

## 6. Question 2 · part c.ii · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL